

Headlamp Adjustment



Special Tool(s)

 ST3041-A	Vision 100 Photometric Headlamp Aimer 196-00005 or equivalent
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Aiming Methods

- NOTE:** Refer to the Owners Literature for the headlamp adjustment screw location.
- NOTE:** Consult your state vehicle inspection manual for recommended tolerance ranges for visual aiming.
- NOTE:** Horizontal aim is not adjustable.

1. Identify the headlamp type. Vehicles are equipped with visually optically aligned left (VOL) or visually optically aligned right (VOR) headlamps. Molded in small letters on the headlamp lens is one of the following:
 - VOL and SAE
 - VOR and SAE
2. Aim the headlamps using one of the following methods:
 - Photometric aimers (preferred method)
 - Visual or screen method aiming (acceptable method)

Photometric Aiming

1. For the photometric aiming procedure, refer to the appropriate photometric headlamp aimer instruction manual.

Screen Method Aiming

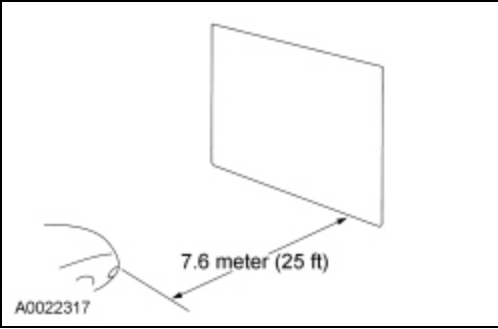
NOTE: [Click here to view a video version of this procedure.](#)

All headlamp types

1. Before starting headlamp adjustment:
 - vehicle must be on level ground.
 - tires must be correctly inflated.
 - vehicle must be normally loaded.
 - headlamps must be clean.
 - air suspension switch must be on (if equipped).
2. **NOTE:** The vertical wall or screen must be a minimum of 2.4 m (8 ft) wide.

NOTE: This procedure should be done in a dark environment to effectively see the headlamp beam pattern.

Park the vehicle on a level surface with the low beam headlamp lens surface approximately 7.6 m (25 ft) from the vertical wall or screen directly in front of it.

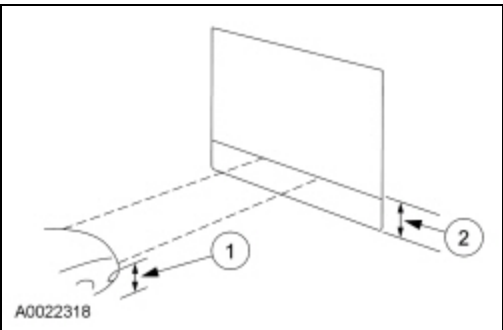


3. **NOTE:** The bulb center of the low beam bulb is sometimes marked on the lens (circle, crosshair or other mark) or is the center of the low beam reflector, bulb shield or the low beam projector inner lens.

Mark a horizontal reference line on the vertical wall or screen.

1. Measure the distance between the center of the low beam headlamp bulb and ground. Record the measurement.
2. Make a 2.4 m (8 ft) horizontal mark (using masking tape) on the vertical wall or screen at the height as follows:

Headlamp bulb center height	Horizontal reference line height
below 95 cm (37.5 inches)	same as bulb center height
95 cm - 105 cm (37.5 inches - 41.5 inches)	bulb center height minus 1.3 cm (.5 inch)
above 105 cm (41.5 inches)	bulb center height minus 2.5 cm (1.0 inch)



4. Turn on the low beam headlamps to illuminate the wall or screen and open the hood.
5. **NOTE:** The cut off of the beam pattern is the horizontal line of the beam pattern where there is **MAXIMUM** change between light & dark

On the wall or screen, locate the cut off of the beam pattern.

VOR type headlamps

NOTE: Procedure applies to both left and right headlamps with "VOR" molded on lens.

6. **NOTE:** The appearance of the VOR beam pattern may vary between vehicles.

There is a distinct cutoff in the right portion of the beam pattern.

1. Horizontal reference line
2. Cut off
3. High intensity zone



7. **NOTE:** *Align one headlamp while covering the other headlamp.*

Align the headlamps to the horizontal reference line

- Adjust the headlamp as necessary using the headlamp adjusting screw.
- Repeat this step for the remaining headlamp.

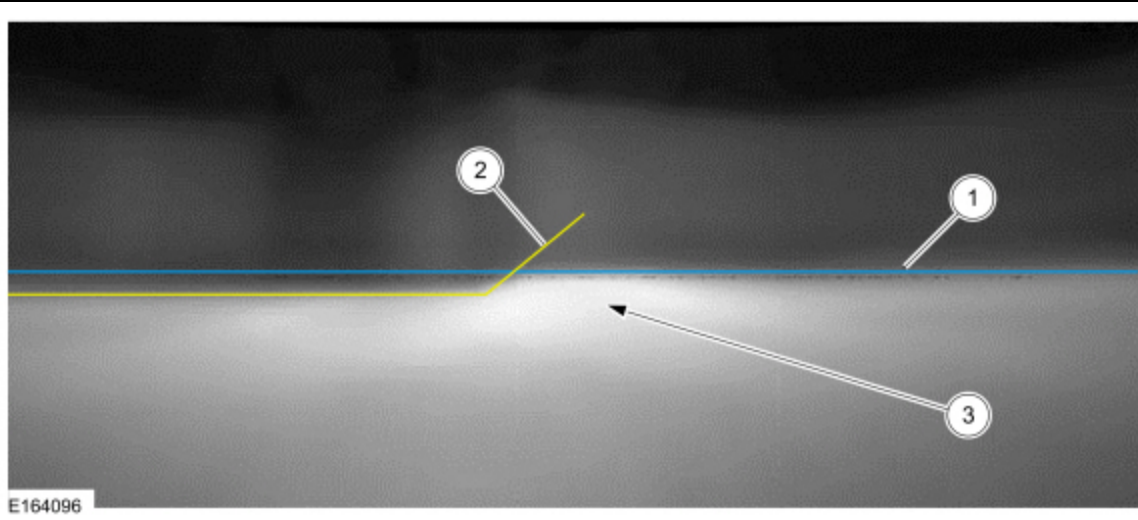
VOL type headlamps

NOTE: *Procedure applies to both left and right headlamps with "VOL" molded on lens.*

8. **NOTE:** *The appearance of the VOL beam pattern may vary between vehicles.*

There is a distinct cutoff in the left portion of the beam pattern. The edge of this cutoff should be positioned 5 cm (2 in) below the horizontal reference line.

1. Horizontal reference line
2. Cut off
3. High intensity zone



9. **NOTE:** *Align one headlamp while covering the other headlamp.*

Align the headlamps to the horizontal reference line

- Adjust the headlamp as necessary using the headlamp adjusting screw.
- Repeat this step for the remaining headlamp.